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Hiring Team
Optiver
Sydney, Australia

Dear Hiring Team,

I am writing to apply for the FPGA Engineer position at Optiver in Sydney. I am an embedded systems and FPGA-oriented engineer with experience in real-time systems, low-level C/C++ development, Linux-based workflows and FPGA-based digital signal processing. Optiver's focus on low-latency execution, network optimisation and high-performance engineering strongly matches the type of systems work that motivates me.

My strongest technical alignment with this role is in translating signal-processing and real-time requirements into hardware-aware implementations. In my master's research, I worked on Digital Predistortion for RF power amplifier linearisation, modelling nonlinear systems in Python and translating the algorithms into fixed-point VHDL architectures for FPGA-oriented implementation. This involved complex arithmetic, delay lines, coefficient quantisation, pipelined datapaths and validation between Python models and VHDL simulation outputs – the kind of detail-oriented work where correctness, latency and resource constraints matter.

Professionally, I have worked as an Embedded & Software Engineer developing low-level C/C++ software for real-time telemetry and connected equipment monitoring systems. This gave me practical experience debugging hardware/software interactions, implementing communication interfaces, working in Linux environments and integrating embedded devices with backend systems. I am comfortable working close to hardware while also using software tooling in Python, Git and Linux to validate, automate and improve engineering workflows.

I am particularly interested in Optiver because the role combines FPGA fundamentals, Linux, C/C++/Python and network-aware low-latency systems. I would be excited to contribute to execution and connectivity platforms where engineering decisions have immediate performance impact, while learning from a team operating at a very high technical standard.

Thank you for considering my application. I would welcome the opportunity to discuss how my FPGA, DSP, embedded systems and Linux background could contribute to Optiver's engineering team in Sydney.

Sincerely,
Leonardo De Andrade Santos